Operating Experience of CFB Boiler Burning Lignite with High Shrinkage Ash Property

Paper Summary

SUMMARY PASSAGE 1

Introduction

With experience gained from successfully operating lignite based 125 MWe plants at Surat in Gujarat, Bharat Heavy Electricals Limited (BHEL) had supplied 2x125 MWe CFB power plants at Barsingsar, Rajasthan for burning lignite with the characteristics mentioned above. Herein is described the design of the CFB boilers installed at Barsingsar, problems occurred during operation and resolution provided for achieving full load.

SUMMARY PASSAGE 2

Barsingsar Lignite Cfb Boiler Design

Bore hole samples of lignite were extracted for determining fuel composition and ash deformation temperature for arriving at basic design parameters. Fuel samples prepared in accordance with ASTM-D 2013 were analyzed at laboratory in BHEL, Tiruchirapalli (Trichy) premises. Ash fusibility tests carried out as per ASTM D-1857 indicated the initial deformation temperature to be 1320Â°C.

SUMMARY PASSAGE 3

Issues Faced While Commissioning

During initial operation of boiler beyond 30% load with lignite feeding, pressure increase above design value at either cyclone to seal pot standpipe, decrease in seal pot blower pressure and a corresponding decrease in inventory of ash (differential pressure -(dp)) in the combustor was observed. Eventually the boiler tripped on account of combustor dp dropping below the minimum required value. On inspecting the primary loop, it was found that circulating material had agglomerated in standpipe of cyclone to loop seal as indicated in figure 2.

SUMMARY PASSAGE 4

Analysis And Discussion

2. High sintering was noticed during compression strength tests in case of lignite ash at 900Â°C whereas cyclone deposit did not undergo any sintering during compression test, however 30 % by weight lignite ash addition to the cyclone deposit resulted in formation of hard sinters at 900Â°C. 3.

SUMMARY PASSAGE 5

Resolution Provided

Adjusting primary and secondary air flow 2. Varying combustor inventory to achieve higher bottom temperature 5.1 Description of process modification Above25% load, with lignite feeding, the combustor inventory was maintained at lower level than design while correspondingly admitting lesser primary air through grate than the original design value for the same load and injecting the rest as secondary air. This resulted in bed temperatures of about 900Â°C and the temperature at combustor top was in the range 730-790Â°C.

SUMMARY PASSAGE 6

Conclusion

As observed from ash tests, the mineralogy of the lignite exhibits varying sintering behaviour for different ash composition although extracted from the same mines. CFB combustion has been proven to be the ideal technology for firing lignite with high ash shrinkage property. Rated full load capacity of 125 MWe was successfully demonstrated at site.